

Integration by parts (pdf 2)

1. $\int x e^x dx$

$u = x \quad v = e^x$

$u' = 1 \quad v dv = e^x$

Formule: $\int u v' dx = u \cdot v - \int (u' \cdot v) dx$

$= x \cdot e^x - \int (1 \cdot e^x) dx$

$= x \cdot e^x - e^x + C$

$e^x(x-1) + C \quad \text{An}$

2. $\int (x \cos x) dx$

$u = x \quad v = \cos x$

$u' = 1 \quad v dv = -\sin x$

$\int u \cdot v' = u \cdot v - \int (u' \cdot v) dx$

$x \cdot \sin x - \int (1 \cdot \sin x) dx$

$x \cdot \sin x + \cos x + C$

~~$\sin x(x+1) + C$~~

3. $\int_0^{\ln 2} (x e^{-3x}) dx$

$u = x \quad v = e^{-3x}$

$du = 1 \quad v dv = \frac{e^{-3x}}{-3}$

$\Rightarrow \int = \frac{x \cdot e^{-3x}}{-3} - \int (-1 \cdot \frac{e^{-3x}}{-3}) dx$

$\left[\frac{-x \cdot e^{-3x}}{3} + \frac{1 \cdot e^{-3x}}{-3} \right]_0^{\ln 2}$

$\left[\frac{-x \cdot e^{-3x}}{3} + \frac{e^{-3x}}{9} \right]_0^{\ln 2}$

$$\left[-\frac{(\ln 2) \cdot e^{-3(\ln 2)}}{3} - \frac{e^{-3(\ln 2)}}{9} \right] - \left[-\frac{1}{9} \right]$$

$$+ \frac{0.6 \cdot 2.07}{3} + \frac{2.07}{9} + \frac{1}{9}$$

$$= \frac{1.842}{3} + \frac{3.07}{9} = \frac{3.726 + 3.07}{9}$$

$$\frac{6.796}{9} \text{ A}$$

$$4. \int_0^{\infty} x e^{-3x} dx$$

$$u = x \quad v = e^{-3x}$$

$$u' = 1, \quad v dv = -\frac{e^{-3x}}{3}$$

$$= \frac{x \cdot e^{-3x}}{3} - \int \left(\frac{-1 \cdot e^{-3x}}{3} \right) dx$$

$$\left[\frac{x \cdot e^{-3x}}{3} + \frac{1}{3} \cdot \frac{e^{-3x}}{3} \right]_0^{\infty}$$

$$\left[\frac{x \cdot e^{-3x}}{3} - \frac{e^{-3x}}{9} \right]_0^{\infty} \text{ A}$$

$$4. \int \ln x \, dx$$

$$u = \ln x$$

$$v = x \, dx$$

$$u' = \frac{1}{x}$$

$$v dv = x$$

$$\int u \cdot v = u \cdot \int v dv - \int (u' \cdot v) dv$$

$$u \cdot v = \int (u \cdot v) dv$$

$$x \ln x = \int \left(\frac{1}{x} \cdot x \right) dx$$

$$x \ln x = x + C$$

$$6. \int \arcsin u \, du$$

$$u = \arcsin u$$

$$v = du$$

$$u' = \frac{1}{\sqrt{1-u^2}}$$

$$v dv = u$$

$$\int u \, v = u \cdot v - \int (u' \cdot v \, dv) \, dv$$

$$u \cdot \arcsin u - \int \left(\frac{u}{\sqrt{1-u^2}} \right) dv$$

$$\int \frac{u}{\sqrt{1-u^2}}$$

$$u = 1-u^2$$

$$u' = -2u$$

$$\int \frac{u}{-2\sqrt{u}} \Rightarrow$$

$$= -\frac{1}{2} \int u^{-\frac{1}{2}} = -\frac{1}{2} \cdot 2\sqrt{u} = -\sqrt{1-u^2}$$

$$= \int u \cdot \arcsin u - \left(-\sqrt{1-u^2} \right) = u \cdot \arcsin u + \sqrt{1-u^2} + C$$

$$7. \int (e^x \sin x) \, dx$$

$$u = \sin x$$

$$v = e^x$$

$$u' = \cos x$$

$$v dv = e^x$$

$$\int u \, v = u \cdot v - \int (u' \cdot v \, dv) \, dv$$

$$= e^x \cdot \sin x - \int (\cos x \cdot e^x) \, dv, u = \cos x, v = e^x$$

$$= e^x \cdot \sin x - \left[\cos x \cdot e^x - \int (-\sin x \cdot e^x) \, dv \right]$$

$$u = -\sin x, u' = -\cos x$$

$$v = e^x, v dv = e^x$$

$$= e^x \cdot \sin x - \left[\cos x \cdot e^x - \left[-\sin x \cdot e^x - \int (-\cos x \cdot e^x) \, dv \right] \right]$$

$$\begin{aligned}
 i &= \sin x \cdot e^x - [e^x \cdot \cos x + \int (\cos x \cdot e^x) dx] \\
 i &= \sin x \cdot e^x - e^x \cdot \cos x - \int (\sin x \cdot e^x) dx \\
 i + \int \sin x \cdot e^x &= e^x (\sin x - \cos x) \\
 2i &= e^x (\sin x - \cos x) \\
 i &= \frac{e^x (\sin x - \cos x) + C}{2} \quad \text{Ans.}
 \end{aligned}$$

$$18. \int \sin x (x^2 \cdot e^{-3x}) dx$$

$$\begin{aligned}
 u &= x^2 & v &= e^{-3x} \\
 du &= 2x & v dv &= -\frac{e^{-3x}}{3}
 \end{aligned}$$

$$i = x^2 \cdot e^{-3x} - \int \left(\frac{-2x \cdot e^{-3x}}{3} \right) dx$$

$$i = x^2 \cdot e^{-3x} + \frac{2}{3} \int (x \cdot e^{-3x}) dx$$

$$\begin{aligned}
 u &= 2x & v &= e^{-3x} \\
 u' &= 1 & v dv &= -\frac{e^{-3x}}{3}
 \end{aligned}$$

$$i = x^2 \cdot e^{-3x} + \frac{2}{3} \left[2x \cdot e^{-3x} - \int \left(\frac{-2 \cdot e^{-3x}}{3} \right) dx \right]$$

$$i = x^2 \cdot e^{-3x} + \frac{2}{3} \left[2x \cdot e^{-3x} + \frac{2}{3} \int e^{-3x} dx \right]$$

$$i = x^2 \cdot e^{-3x} + \frac{2}{3} \left[2x \cdot e^{-3x} + \frac{2}{9} e^{-3x} \right] + C$$

$$i = x^2 \cdot e^{-3x} + 4x \cdot e^{-3x} + \frac{4}{9} e^{-3x} + C$$

Ans

$$1. \int x e^{2x} dx$$

$$u = x$$

$$v = e^{2x}$$

$$u' = 1$$

$$v dv = \frac{e^{2x}}{2}$$

$$\int u \cdot v = u \cdot v - \int (u' \cdot v) dv$$

$$i = x \cdot e^{2x} - \int \left(1 \cdot \frac{e^{2x}}{2}\right) dv$$

$$i = \frac{e^{2x} \cdot x}{2} - \frac{1}{2} \cdot \int (e^{2x}) dv$$

$$i = \frac{e^{2x} \cdot x}{2} - \frac{1}{2} \cdot \frac{e^{2x}}{2}$$

$$\left(i = \left(\frac{e^{2x} \cdot x}{2} - \frac{e^{2x}}{4} \right) \right) \text{ Ans.}$$

$$2. \int (x e^{-3x}) dx$$

$$u = x$$

$$v = e^{-3x}$$

$$u' = 1$$

$$v dv = \frac{e^{-3x}}{3}$$

$$= \frac{x \cdot e^{-3x}}{3} - \int \left(1 \cdot \frac{e^{-3x}}{3}\right) dv$$

$$= \frac{x \cdot e^{-3x}}{3} - \frac{e^{-3x}}{3 \cdot 3}$$

$$\frac{e^{-3x} \cdot x}{3} - \frac{e^{-3x}}{9} \text{ Ans.}$$

$$5. \int x 2^x dx$$

$$u = x$$

$$v = 2^x$$

$$u' = 1$$

$$v dv = \frac{2^x}{\ln 2}$$

$$\int u \cdot v = u \cdot v - \int (u') \cdot v \, dv$$

$$\frac{n \cdot 2^n}{\ln 2} - \int \left(\frac{1 \cdot 2^n}{\ln 2} \right) dv$$

$$\frac{2^n \cdot n}{\ln 2} - \frac{2^n}{\ln 2} + C \text{ Ans}$$

$$6. \int n^2 2^n \, dn$$

$$u = n^2, \quad v = 2^n$$

$$u' = 2n \quad v \, dv = \frac{2^n}{\ln 2}, \quad \therefore n' = 1$$

$$i = \frac{2^n n^2}{\ln 2} - \int \left(\frac{2n \cdot 2^n}{\ln 2} \right) dv$$

$$i = \frac{2^n n^2}{\ln 2} - \frac{2}{\ln 2} \left[\int (n \cdot 2^n) \, dn \right]$$

$$i = \frac{2^n n^2}{\ln 2} - \frac{2}{\ln 2} \left[\frac{n \cdot 2^n}{\ln 2} - \int \left(\frac{2^n}{\ln 2} \right) \, dn \right]$$

$$i = \frac{2^n n^2}{\ln 2} - \frac{2}{\ln 2} \left[\frac{2^n n}{\ln 2} - \frac{1}{\ln 2} \cdot \frac{2^n}{\ln 2} \right] + C$$

$$i = \frac{2^n n^2}{\ln 2} - \frac{2}{\ln 2} \left(\frac{2^n n}{\ln 2} - \frac{2^n}{(\ln 2)^2} \right) + C$$

$$\frac{2^n n^2}{\ln 2} - \frac{2^{n+1} n}{(\ln 2)^2} - \frac{2^{n+1}}{(\ln 2)^3} + C$$

Ans

$$9. \int x \ln x \, dx$$

$$u = \ln x$$

$$v = x$$

$$u' = \frac{1}{x}$$

$$v dv = \frac{x^2}{2}$$

$$\frac{\ln x \cdot x^2}{2} - \int \left(\frac{1}{x} \cdot \frac{x^2}{2} \right) dx$$

$$\frac{\ln x \cdot x^2}{2} - \int \left(\frac{x}{2} \right) dx$$

$$\frac{\ln x \cdot x^2}{2} - \frac{1 \cdot x^2}{2 \cdot 2} + C$$

$$\frac{\ln x \cdot x^2}{2} - \frac{x^2}{4} + C \text{ Ans.}$$

$$10. \int_1^9 \frac{\ln x}{\sqrt{x}} \, dx$$

$$u = \ln x$$

$$v = \cancel{x}^{-\frac{1}{2}} x^{-\frac{1}{2}}$$

$$u' = \frac{1}{x}$$

$$v dv = 2\sqrt{x}$$

$$i = \ln x \cdot 2\sqrt{x} - \int \left(\frac{1}{x} \cdot 2\sqrt{x} \right) dx$$

$$\left[\ln x \cdot 2\sqrt{x} - \frac{1}{2} \left(\int \sqrt{x} \, dx \right) \right]_1^9$$

$$\left[\ln x \cdot 2\sqrt{x} - \frac{1}{2} \cdot \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^9$$

$$\left[\ln x \cdot 2\sqrt{x} - \frac{2x^{\frac{3}{2}}}{6} \right]_1^9 \quad \text{Ans.}$$

$$\left[\ln 9 \cdot 2\sqrt{9} - \frac{2(9)^{\frac{3}{2}}}{6} \right] - \left[\ln 1 \cdot 2\sqrt{1} - \frac{2(1)^{\frac{3}{2}}}{6} \right]$$

$$2 \cdot 2 \cdot 6(13.2 - 9) - \left(\frac{1}{3} \right)$$

$$4 \cdot 2 - \frac{1}{3} \Rightarrow 3.86 \text{ Ans.}$$